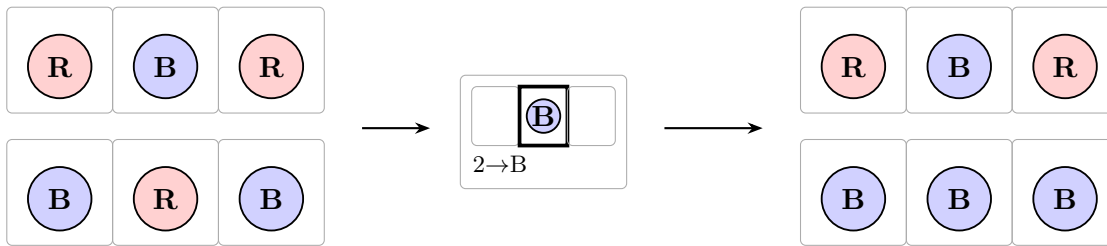
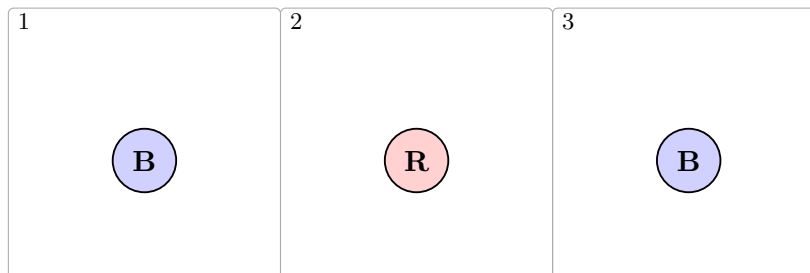
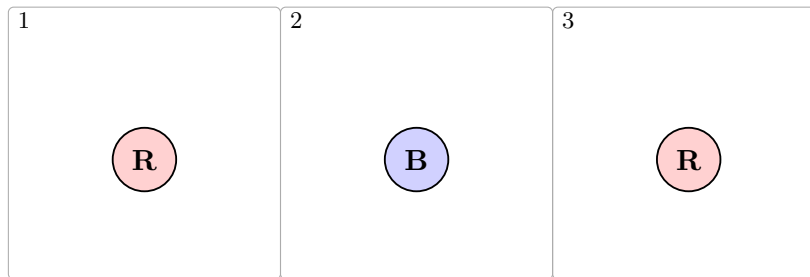


Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.

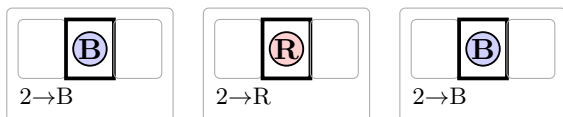
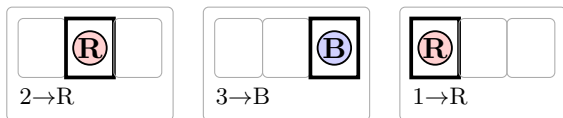
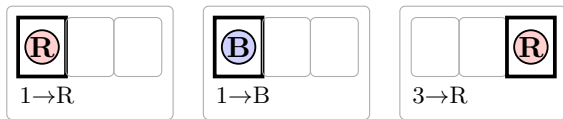


Problem 1: Find the fewest instructions needed to make these boards match. Build pairs needing fewer instructions. Can any pair need more?



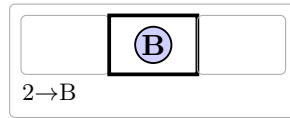
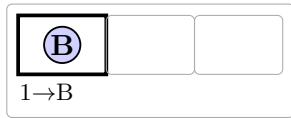
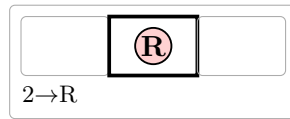
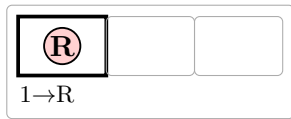
Put a small marker above a slot when an instruction first names it. Leave the marker there for the rest of the story.

Problem 2: Which stories guarantee that any two starting boards will finish matching? Explain why your answer holds for every starting pair.



Problem 3: Can some starting pairs match before all three slots have markers? Make one example that does and one that does not.

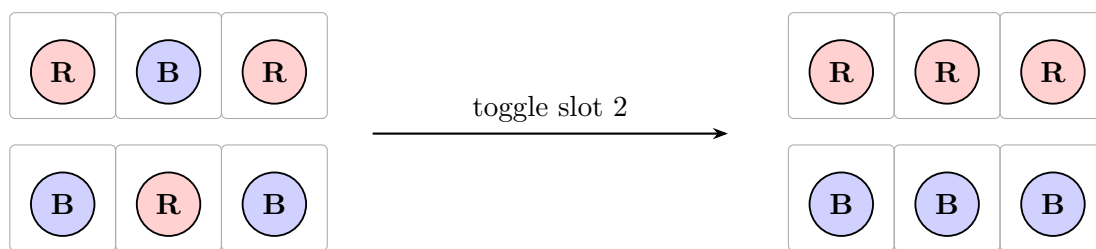
Mix six instruction cards in a cup. Draw, use the instruction on both boards, replace, and mix.



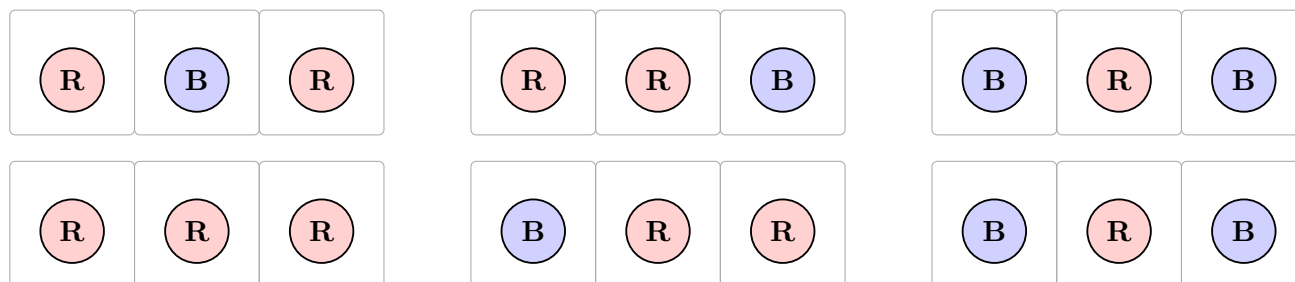
Problem 4: A friend says three draws must make every pair match. Make a three-draw story that tests that claim. Could even ten draws fail?

Problem 5: Find the shortest instruction story that guarantees a match for every starting pair. Explain why no shorter story can give that guarantee.

A toggle changes red to blue or blue to red in its slot, on both boards. It never sets a chosen color.



Problem 6: Use only shared toggles on these starting pairs. Which can be made to match, and why?



Problem 7: What changes if a story may use both color-setting instructions and toggles? Invent a story that makes every starting pair match, and a story that fails for at least one pair. Show a pair where it fails.